

The Anatomy of Firm Scope^{*}

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Abstract

I show that firm scope operates along two economically distinct dimensions. Text-based product-market clusters measure related scope—closely linked sub-markets sharing a technological substrate—while the Compustat segment count measures distant scope—operationally separable lines of business. Using a panel of U.S. public firms over 1990–2019, I estimate local projections on patent-value innovation shocks and find that these dimensions govern opposite regimes. Broad related scope generates synergy: firms tilt intangible investment toward R&D, attract rival entry, and convert innovation into persistent productivity and markup gains. High distant scope generates friction: rents dissipate, productivity gains vanish, and intangible accumulation shifts toward brand and organisational capital. The superstar-firm profile emerges from the structure, not the level, of scope.

Keywords: firm scope, diversification, intangibles, productivity, markups

JEL Classification: L25, O31, E22.

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1. Introduction

Modern firms rarely confine themselves to a single product market, yet the structure of their scope varies sharply. Some firms operate across many closely related sub-markets that draw on a common technological base. Others span operationally distant business domains that share little beyond common ownership. The distinction between related and unrelated diversification has shaped corporate research since [Panzar and Willig \(1981\)](#) and [Teece \(1980\)](#), and it remains central to work on innovation synergies ([Bena and Li, 2014](#)) and product-market relatedness ([Hoberg and Phillips, 2010](#); [Hoberg and Phillips, 2016](#)). This paper asks how the structure of firm scope, rather than its extent, shapes productivity, markups, the composition of intangible investment, and exposure to competitive threat.

Throughout the paper I distinguish two dimensions of scope. *Related scope* is the number of closely related sub-markets a firm serves within the boundary of a shared technological base. These sub-markets draw on common scientific knowledge, production know-how, and complementary assets, so they lie close to one another in product space. *Distant scope* is the number of operationally distinct business domains a firm spans beyond that boundary. These domains share neither a technological substrate nor a customer base, and firms typically organise them as separate divisions with their own management. The two dimensions are conceptually independent. A firm can be broad in one and narrow in the other, and the central claim of the paper is that the two configurations place firms in opposite economic regimes.

I measure each dimension directly. Related scope is captured by the text-based product-market clusters of [Hoberg and Phillips \(2025\)](#), which resolve the product descriptions in a firm’s 10-K filings into fine-grained markets that lie close to one another in product space. Distant scope is captured by the number of business segments a firm reports in the Compustat Historical Segment file. Segments are disclosed under SFAS 131 precisely to partition activity into operationally distinct lines of business, so the segment count records how many largely non-overlapping domains a firm spans. Tesla illustrates the

difference. Hoberg and Phillips resolve Tesla into eleven product-market clusters, covering automotive subsystems (brakes, trim, axles, chassis), automotive safety, car dealerships, batteries, solar and renewable energy, utility-scale electric power, smart metering, power electronics, hardware and software solutions, and ticketing systems, all anchored in a common electrochemical and energy-systems substrate. Compustat aggregates the same firm into two operationally distant segments, Automotive and Energy Generation & Storage.¹ The two datasets are not rival proxies for one object. They are complementary windows on different layers of scope.

The economic content of the distinction lies in technological proximity. When a firm expands into closely related sub-markets, it remains within the boundary of a common technological base, and proximity makes knowledge portable. A patent, a production process, or a body of engineering know-how developed for one sub-market can be redeployed in neighbouring sub-markets at little additional cost, so the returns to R&D compound with related scope ([Panzar and Willig, 1981](#); [Bena and Li, 2014](#)). When a firm expands into operationally distant domains, technological proximity no longer holds, and two different forces govern the economics of scope. The first is a span-of-control constraint. Managerial attention and coordination capacity must stretch across domains that share no common substrate, so each additional domain dilutes oversight of the others, in the spirit of [Penrose \(1959\)](#) and [Lucas \(1978\)](#). The second is the growing weight of headquarters intangibles. Brand value and organisational capital attach to the firm rather than to any particular technology ([Prescott and Visscher, 1980](#); [Eisfeldt and Papanikolaou, 2013](#)), and evidence on intra-firm flows confirms that what moves across a firm's establishments is precisely this kind of intangible input rather than physical goods ([Atalay et al., 2014](#)). [Keskin \(2026\)](#) studies this margin in isolation, documenting that firms with many distant segments reallocate intangible investment away from R&D and toward brand and organisational capital, and formalising the mechanism in a Schumpeterian model with span-of-control frictions. This paper separates the two dimensions of scope within the same sample and shows that they place firms in opposite regimes. Related scope is a regime

¹See Table [A1](#) for details.

of synergy, in which innovation rents compound across sub-markets. Distant scope is a regime of friction, in which those rents dissipate and the intangible portfolio tilts toward brand and organizational capital investments.

I take these predictions to firm-level Compustat data for fiscal years 1990 to 2019. To the two scope measures I add the product-market fluidity index of [Hoberg *et al.* \(2014\)](#), which captures how quickly rivals encroach on a firm's product space, together with firm-level measures of total factor productivity, markups, and the composition of intangible capital. Following the taxonomy of [Keskin \(2026\)](#), I separate intangible investment into R&D on the one hand and brand value and organisational capital on the other.² The empirical design follows the local-projection method of [Jordà \(2005\)](#), applied to the patent-value shocks of [Kogan *et al.* \(2017\)](#), which measure the economic value of a firm's patents from the stock-market reaction to their grant. The design traces the dynamic response of each outcome to a well-identified innovation shock and allows the response to differ along the two dimensions of scope.

The synergy and friction predictions are borne out. After a patent-value shock, firms with broad related scope absorb innovation aggressively. They accelerate investment in R&D, attract rival entry into their product space, and convert the shock into persistent gains in productivity and markups. Firms spread across many operationally distant segments respond weakly on every margin, their productivity response is indistinguishable from zero, and their intangible investment tilts toward brand and organisational capital rather than research. Innovation rents are appropriated where related scope supplies synergistic deployment outlets. They dissipate where activity is fragmented across domains in which R&D does not transfer and managerial control is stretched thin.

The combined message is that *scope is not a scalar*. Two firms with the same number of markets can inhabit opposite economic regimes depending on whether those markets lie within a shared technological base or beyond it. Firms that combine broad related scope

²[Keskin \(2026\)](#) labels these two categories transferable and embedded intangibles: R&D is transferable because its returns travel across technologically related markets, whereas brand value and organisational capital are embedded in the firm itself and remain deployable across all of its lines of business.

with few distant segments exhibit the profile of high productivity and persistent markups that [Autor *et al.* \(2020\)](#) and [De Loecker *et al.* \(2020\)](#) attribute to superstar firms and the rise of market power. Those studies characterise superstar firms by their outcomes rather than by their scope, and the structure of scope offers an account of what produces the profile. A shared technological base lets R&D compound across synergistic sub-markets, raising productivity, sustaining markups, and concentrating intangible accumulation in the input whose returns the firm can appropriate across its full product-market portfolio. Firms that combine many distant segments with narrow related scope display the configuration of the diversified conglomerates studied by [Lang and Stulz \(1994\)](#), [Berger and Ofek \(1995\)](#), and [Schoar \(2002\)](#). In these firms the absence of a shared base dissipates the appropriability of R&D, span-of-control frictions bind, and brand and organisational capital become the residual intangibles that bridge the firm's domains, compressing productivity and markups. The familiar cross-firm gradients in productivity, market power, and intangible composition are therefore joint consequences of the underlying structure of scope rather than independent margins of firm behaviour.

Related literature. The paper speaks to four bodies of work. The first is the classical literature on economies of scope and corporate diversification ([Panzar and Willig, 1981](#); [Teece, 1980](#); [Helfat and Raubitschek, 2000](#); [Lang and Stulz, 1994](#); [Berger and Ofek, 1995](#); [Schoar, 2002](#); [Maksimovic and Phillips, 2002](#); [Villalonga, 2004](#); [Custódio, 2014](#)), together with the span-of-control tradition of [Penrose \(1959\)](#) and [Lucas \(1978\)](#). I show that the diversification discount is a feature of distant scope, that related scope generates the opposite pattern, and that the two effects coexist within the same firm and the same sample once the dimensions are separated. The second is the literature on rising markups and superstar firms ([De Loecker *et al.*, 2020](#); [Autor *et al.*, 2020](#); [Crouzet and Eberly, 2019](#); [Crouzet and Eberly, 2023](#)). The joint movement of markups and productivity at the core of recent concentration narratives is driven by firms with broad related scope and few distant segments. The third is the literature on intangible capital ([Corrado *et al.*, 2009](#); [Prescott and Visscher, 1980](#); [Eisfeldt and Papanikolaou, 2013](#); [Eisfeldt and Papanikolaou, 2014](#); [Pe-](#)

ters and Taylor, 2017; Haskel and Westlake, 2017; Keskin, 2026). The structure of scope emerges as a first-order determinant of intangible composition: firms tilt toward R&D when scope is related and toward brand and organisational capital when scope is distant, because only the latter travel across operationally distant domains. The fourth is the empirical innovation literature built on patent-value shocks (Kogan *et al.*, 2017; Bena and Li, 2014). The appropriability of innovation depends on the structure of scope rather than on its level. Related scope amplifies innovation rents and distant scope dissipates them.

Paper organisation. The remainder of the paper proceeds as follows. Section 2 describes the data sources and presents summary statistics on the two scope measures. Section 3 details the firm-level measurement of productivity, markups, and the components of intangible capital. Section 4 estimates the dynamic responses to patent-value innovation shocks across the two scope dimensions. Section 5 concludes.

2. Data and Summary Statistics

The analysis combines five firm-level data sources. Compustat Fundamentals supplies the financial variables from which productivity, markups, and the composition of intangible investment are constructed. The Compustat Historical Segment file supplies the measure of distant scope, and the Hoberg–Phillips text-based scope dataset supplies the measure of related scope. The fluidity index of Hoberg *et al.* (2014) measures the competitive threat each firm faces, and the patent-value dataset of Kogan *et al.* (2017) provides the innovation shocks that drive the dynamic analysis. All sources are merged at the firm-year level, and the sample covers fiscal years 1990 to 2019. This section describes each source in turn and then documents the cross-sectional behaviour of the two scope measures, which is itself informative about the structure of firm scope.

Firm-level financial data. Compustat Fundamentals provides comprehensive financial information for publicly listed U.S. firms, with longitudinal coverage of balance-sheet

items, income-statement components, and cash-flow data, and with firm identifiers that permit clean linkage to the scope, fluidity, and patent datasets. I use Compustat to construct the firm-level measures of markups, productivity, and intangible composition described in Section 3, and as the spine onto which the remaining datasets are merged. The sample excludes the finance and utilities sectors, whose regulated pricing and distinct accounting conventions make production functions and markups incomparable with the rest of the economy, and drops firm-years with missing or non-positive R&D, SG&A, employment, or sales. The restriction to positive R&D and SG&A is substantive rather than cosmetic: the composition of intangible investment is defined as the ratio of R&D spending to spending on brand and organisational capital, and it is only meaningful for firms that invest in both.

Distant scope. Measuring distant scope, the operationally separable lines of business a firm spans, requires data on the broad units that firms themselves treat as substantively distinct. I obtain this from the Compustat Historical Segment file, which compiles firms' mandated segment-level disclosures from 1976 to the present. From 1976 to 1997 these disclosures were governed by SFAS No. 14, under which segments were defined along industry lines. From fiscal year 1998 onward they are governed by SFAS No. 131, codified as ASC 280 in 2009, which replaced the industry approach with a *management approach*: reportable segments must correspond to the components of the enterprise whose operating results the chief operating decision maker regularly reviews to allocate resources and assess performance. The standard imposes quantitative thresholds, so that a component becomes reportable when it accounts for at least ten percent of the firm's revenue, profits, or assets, and it permits aggregation of components only when they share similar economic characteristics. Distinct reported segments are therefore, by regulatory construction, units that the firm's own management runs and evaluates separately. Compustat assigns a unique identifier (*sid*) to each reported segment, and I define distant scope as

the number of distinct business segments a firm reports in a given year.³ In practice this yields broad partitions that distinguish operationally and informationally separable lines of business rather than fine-grained product variety, so the segment count captures diversification across domains rather than breadth within them. Year fixed effects in every specification absorb aggregate shifts in reporting practice, including the 1998 transition between regimes.

Related scope. Measuring related scope, the fine-grained product-market clusters a firm spans within its operational domain, requires data of a different kind, constructed from what firms sell rather than from how managers report. I use the Hoberg–Phillips text-based scope dataset ([Hoberg and Phillips, 2025](#)), which applies natural-language processing to the business descriptions firms file in their annual 10-Ks. The construction embeds each firm’s product description in a high-dimensional vector-space representation of the U.S. product market and assigns the firm to a set of product-market clusters according to its proximity to other firms in that space. The variable `d2vscope` reports the number of distinct clusters a firm occupies in a given year, and I use it as the measure of related scope throughout. The text-based construction has two advantages for present purposes. First, it is independent of managerial reporting discretion and of static industry classifications, updating annually as firms reposition their products. Second, because clusters are recovered from product-description language, they are typically much narrower than Compustat segments and, for a given firm, lie close to one another in product space. The cluster count therefore captures exactly the within-domain product variety that the segment count is constructed to abstract away from. Table [A1](#) illustrates the contrast for Tesla, Inc., which Compustat aggregates into two distant segments while Hoberg–Phillips resolves into eleven related product-market clusters.

³The Historical Segment dataset reports several segment classifications at the firm-year level. I retain business segments and exclude geographic and operating classifications, as well as residual corporate or eliminations segments, since these do not correspond to lines of business. Beginning in 2015, the WRDS Historical Segment files additionally provide product–service (PD_SRVC) information, which directly identifies the products and services associated with each segment. To construct a consistent measure, I use business segments prior to 2015 and PD_SRVC-based definitions thereafter. Firms without segment records in a given year are excluded from the sample.

Competitive threat. To capture the competitive pressure firms face, I use the product-market fluidity metric of [Hoberg *et al.* \(2014\)](#). Fluidity measures the overlap between a firm’s own product vocabulary and the year-over-year *change* in the vocabulary used by all rival firms, so it rises when competitors are actively moving into the product space the firm occupies. A high fluidity score therefore indicates that rivals are rapidly reconfiguring their offerings around the firm’s position, and it serves in the analysis as a direct measure of the competitive threat surrounding a firm’s product market.

Table 1. Summary Statistics

Variable	N	Mean	SD	P25	Median	P75
Sale	61,104	2,068,272	11,797,105	19,109	98,265	595,534
Emp	61,104	6,104	26,107	119	482	2,691
AT	61,101	2,728,453	18,443,129	22,153	110,624	677,513
XRD	61,104	100,848	628,530	1,361	6,282	27,861
XSGA	61,104	405,702	1,991,132	10,240	36,839	156,267
Productivity	55,845	19.781	23.080	7.706	10.879	22.907
Markup	55,316	1.537	1.030	0.987	1.212	1.663
Competitive Threat	50,285	6.769	3.203	4.437	6.365	8.603

Notes: This table reports summary statistics for sample s1, restricted to firm-years with positive sales, employment, R&D, and SG&A.

Innovation shocks. To trace how firm outcomes evolve around innovation events, I use the patent-value dataset of [Kogan *et al.* \(2017\)](#), which provides patent identifiers, filing and grant dates, firm identifiers, forward citations, and a market-based estimate of each patent’s value. The value measure is constructed from the firm’s idiosyncratic stock return within a narrow event window around the patent grant date, filtered of market movements and converted into a dollar magnitude. Because the exact grant date is set by the Patent Office, the short-window design isolates the surprise component of the market’s valuation of the grant, abstracting from anticipation effects and concurrent firm-level news. Patent-level values are linked to Compustat through the CRSP–Compustat merged identifiers and aggregated to a firm-year innovation shock, whose precise construction is

given in Section 4. This shock provides the plausibly exogenous innovation variation on which the local-projection estimates are built.

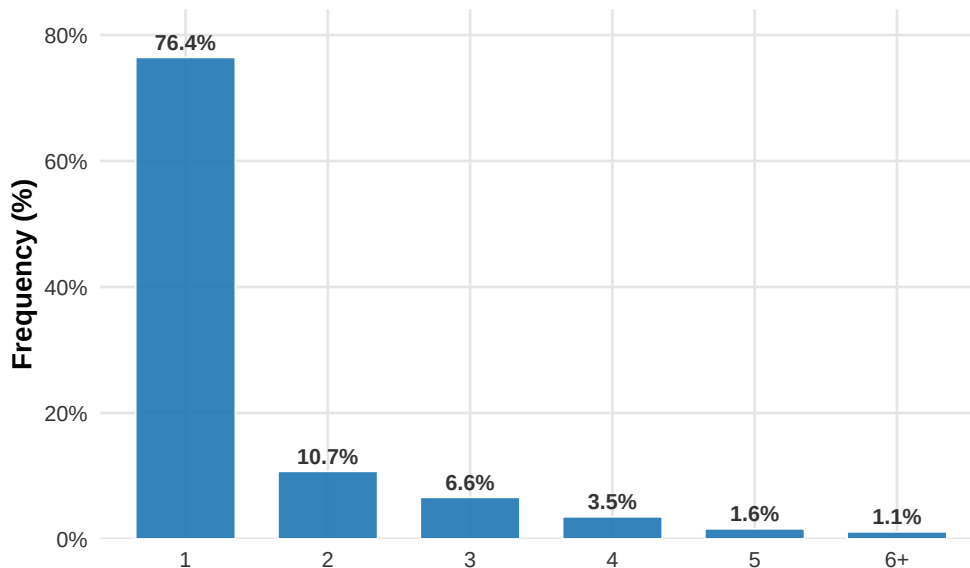


Figure 1. Firm Distribution across Scope (Compustat)

Note: The figure shows the distribution of firm-year observations across the number of reported business segments (*sid_count*). Observations with zero segments are excluded. Values greater than five are grouped into the “6+” category.

Summary statistics. Table 1 reports pooled summary statistics for the main variables over fiscal years 1990–2019. The underlying Compustat sample exceeds 70,000 firm-year observations on the raw size measures (sales, employment, total assets, R&D, and SG&A) and displays the pronounced right-skew familiar from the Compustat cross-section: median sales and total assets are roughly two orders of magnitude smaller than their means, and all size variables accordingly enter the analysis in logarithms. Firm-level productivity and markups are estimated with the method of [Olley and Pakes \(1996\)](#), whose implementation is detailed in Section 3. Competitive threat, measured by the [Hoberg *et al.* \(2014\)](#) fluidity index, has a mean of 6.78 and a standard deviation of 3.21, in line with the moments reported in prior work. Because the two scope datasets cover overlapping but non-identical subsets of Compustat, the merged estimation samples are smaller than the fundamentals panel. Appendix Table A4 reports summary statistics separately for the

distant-scope and related-scope merged samples and confirms that the merges do not distort the distributions of size, productivity, or markups relative to the underlying Compustat cross-section

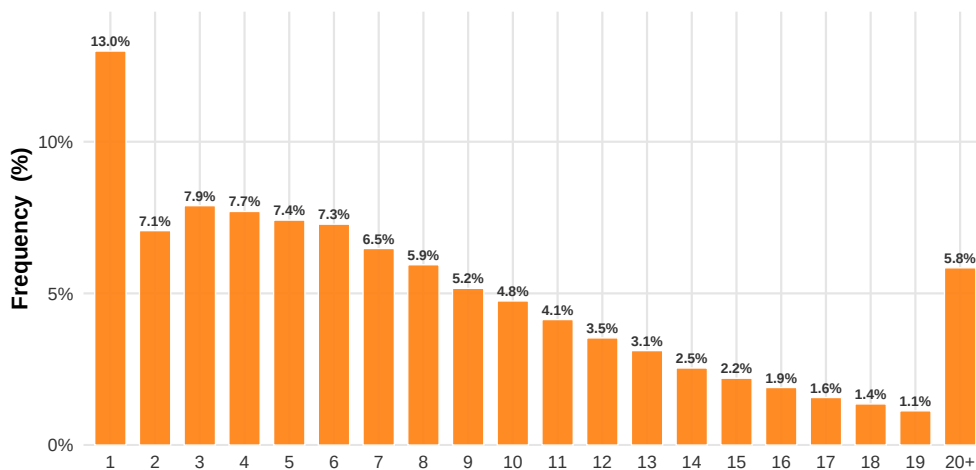


Figure 2. Firm Distribution across Scope (Hoberg-Phillips)

Note: The figure shows the distribution of firm-year observations across the firm scope measure (*d2vscope*), which captures the breadth of a firm's product market activity. Observations with a scope value of zero are excluded. Values greater than nineteen are grouped into the "20+" category.

The two margins of scope. Figures 1 and 2 plot the pooled firm-year distributions of the two scope measures, and the contrast between them is the first substantive fact of the paper. Distant scope is heavily concentrated at a single segment: 76.4% of firm-year observations report one business segment, 10.7% report two, and the share declines sharply thereafter, with only 6.6% of observations reporting four or more segments. Diversification across operationally separable lines of business is, in the panel, the exception rather than the rule, confined to a narrow upper tail. Related scope, in contrast, is widely dispersed. The modal firm-year occupies a single cluster, but the mode accounts for only 13% of observations. Mass is spread smoothly across the interior of the support, with roughly 59% of observations falling between two and ten clusters and a non-trivial right tail in which 5.8% of observations occupy twenty or more. The same population of firms is therefore narrow along one dimension and broad along the other: firms rarely straddle operationally distant domains, yet within their domain they routinely spread across

many neighbouring product markets. This is precisely the pattern the introduction anticipates. The two measures record different margins of expansion rather than the same margin at different resolutions, and collapsing them into a single count would average a dimension along which almost all firms are concentrated with one along which they are widely dispersed.

3. Measurement

The dynamic analysis requires firm-level measures of total factor productivity, markups, and the composition of intangible investment. This section describes how each is constructed from the Compustat data.

Production function and productivity. I estimate firm-level total factor productivity with the two-stage proxy-variable approach of [Olley and Pakes \(1996\)](#), separately for each two-digit NAICS industry. Consider the gross-output production function

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \varepsilon_{it}, \quad (1)$$

where y_{it} , l_{it} , k_{it} , and m_{it} denote log sales, log employment, log capital stock, and log intermediate inputs for firm i in year t ,⁴ ω_{it} is a productivity state observed by the firm before it chooses inputs, and ε_{it} is an i.i.d. disturbance uncorrelated with input choices. The estimation sample is restricted to observations satisfying $\text{sale} > 0$, $\text{ppent} > 0$, $\text{cogs} > 0$, $\text{emp} > 0$, $\text{capx} > 0$, and $\text{cogs} < \text{sale}$, where the final restriction ensures positive gross profits. The restriction to strictly positive capital expenditure also serves the estimator, since the investment inversion below is defined only where investment is strictly positive. All nominal variables are deflated to real terms prior to estimation: output by the GDP deflator, intermediate inputs by the Producer Price Index, and the capital stock

⁴The Compustat counterparts are SALE (sales), EMP (employment), PPENT (net property, plant, and equipment), and COGS (intermediate inputs).

and capital expenditure by the Gross Private Domestic Investment deflator.⁵ The central identification problem is the classic simultaneity of [Marschak and Andrews \(1944\)](#): the firm observes ω_{it} when choosing its inputs, so input choices are correlated with the productivity term and OLS estimates of (1) are biased. Identification rests on the timing of input choices. Capital and labour are quasi-fixed state variables: capital is determined by investment decided in period $t - 1$, and employment is subject to hiring and firing frictions, so neither can respond to the current productivity innovation. Intermediate inputs are the flexible input and adjust to ω_{it} within the period.

Markups. Firm-level markups follow the production approach of [De Loecker *et al.* \(2020\)](#), which requires no assumptions on demand or on the mode of competition. Cost minimisation implies that, for any input free of adjustment costs, the output elasticity of that input equals the markup multiplied by the input's expenditure share in revenue. Rearranging gives

$$\sigma_{it} = \frac{\hat{\beta}_m^{j(i)}}{s_{it}^m}, \quad (2)$$

where $\hat{\beta}_m^{j(i)}$ is the intermediate-input elasticity estimated for firm i 's industry in (??) and s_{it}^m is the firm's expenditure on intermediate inputs divided by its sales revenue. Intermediates are the appropriate variable input in these data because they adjust freely at annual frequency, whereas employment is subject to hiring and firing frictions that violate the static first-order condition. Within an industry, markup variation is therefore driven by variation in the revenue share of intermediates, while cross-industry variation reflects differences in estimated elasticities as well.

Intangible investment and its composition. Following [Peters and Taylor \(2017\)](#), I measure R&D investment as total R&D expenditure (XRD), and investment in brand value and organisational capital as 30% of Selling, General, and Administrative expenses net of R&D

⁵GDP deflator: <https://fred.stlouisfed.org/series/A191RD3A086NBEA>. PPI: <https://fred.stlouisfed.org/series/WPUID61>. Investment deflator: <https://fred.stlouisfed.org/series/A006RD3A086NBEA>.

(XSGA – XRD).⁶ SG&A encompasses a broad range of expenditures, including advertising, employee compensation, and general operating costs. Once R&D is netted out, most of what remains is routine operating expense consumed within the period, but a portion accumulates inside the firm as lasting intangible capital: advertising builds brand recognition, and the compensation of managers and skilled employees builds organisational knowledge and routines. The 30% fraction isolates this accumulating component, the firm-specific intangible capital composed of brand value and organisational capital, and treats the remainder as operating cost that generates no lasting value.⁷ The composition measure used throughout the paper is the ratio of R&D investment to investment in brand and organisational capital, so that an increase in the ratio indicates a tilt of the firm's intangible portfolio toward research.

4. Dynamic Response to Innovation Shocks

The introduction argued that the two dimensions of scope place firms in opposite regimes with respect to innovation. Related scope supplies deployment outlets: a technological gain originating in one product-market cluster can be redeployed across neighbouring clusters that share the same technological base, so broad related scope enlarges the appropriation channel attached to any given innovation. Distant scope supplies frictions: operationally separable segments fragment managerial attention, technical knowledge does not transfer across them, and the intangibles that do travel, brand and organisational capital, defend existing positions rather than convert innovation into efficiency. Two predictions follow. First, firms with low distant scope should convert an innovation shock into R&D, productivity, and markups more strongly than firms with high dis-

⁶Peters and Taylor (2017) refer to the R&D-based component as knowledge capital and to the SG&A-based component as organisation capital.

⁷Brand value and organisational capital could in principle be identified separately, since advertising expenditure is reported as a distinct line item (XAD) in Compustat. XAD is sparsely reported, however, and conditioning on non-missing values would disproportionately retain firms with large advertising budgets, introducing a systematic selection toward advertising-intensive firms. I therefore treat brand value and organisational capital as a single category.

tant scope. Second, firms with broad related scope should convert the same shock more strongly than firms with narrow related scope. This section tests both predictions.

I estimate impulse responses with local projections (Jordà, 2005), which recover horizon-specific responses without imposing a parametric model of the outcome dynamics and accommodate group heterogeneity directly. The innovation shock is the firm-year aggregate of the Kogan *et al.* (2017) patent values, $V_{it} = \sum_{p \in \mathcal{P}_{it}} \text{PatentValue}_{p,t}$, where $\mathcal{P}_{it}$ is the set of patents granted to firm i in year t , entering as $\log(1 + V_{it})$ to retain firm-years with zero grants. Two features support a causal reading. Grant dates are set by the Patent Office review process, so their timing is predetermined relative to the firm’s contemporaneous strategic decisions, and the value measure isolates the market surprise within a narrow window around the grant, purged of anticipation and concurrent firm news. Conditional on lagged outcomes and firm controls, the shock is therefore plausibly orthogonal to the disturbance. The specification is

$$\Delta_h Y_{ijt} = \alpha_h + \beta_{\text{High}}^h V_{it} D_{it} + \beta_{\text{Low}}^h V_{it} (1 - D_{it}) + \mathbf{\Gamma}_h^\top \mathbf{X}_{i,t-1} + \delta_h Y_{i,t-1} + \theta_j + \lambda_t + u_{ijt}, \quad (3)$$

where $\Delta_h Y_{ijt} \equiv Y_{i,t+h} - Y_{it}$ is the h -period log change in outcome Y for firm i in industry j , estimated at horizons $h = 0, 1, 2, 3$, and $D_{it} \in \{0, 1\}$ indicates the high-scope group along one of the two dimensions. The vector $\mathbf{X}_{i,t-1}$ collects lagged firm-level controls (sales, R&D, and SG&A), $Y_{i,t-1}$ absorbs the pre-shock level of the outcome, and θ_j and λ_t denote two-digit-industry and year fixed effects. Standard errors are clustered at the firm level. I estimate the specification separately for each scope dimension on the corresponding merged sample of Section 2. For the distant-scope split, $D_{it} = 1$ identifies multi-segment firms and $D_{it} = 0$ single-segment firms. A median split is uninformative along this dimension, because the median firm-year reports exactly one segment. For the related-scope split, $D_{it} = 1$ identifies firms whose `d2vscope` lies at or above the sample median and $D_{it} = 0$ firms below it. Scope status is fixed at each firm’s pre-shock classification and held constant over the estimation window, so β_{High}^h and β_{Low}^h trace the responses of two predetermined groups of firms to the same shock.

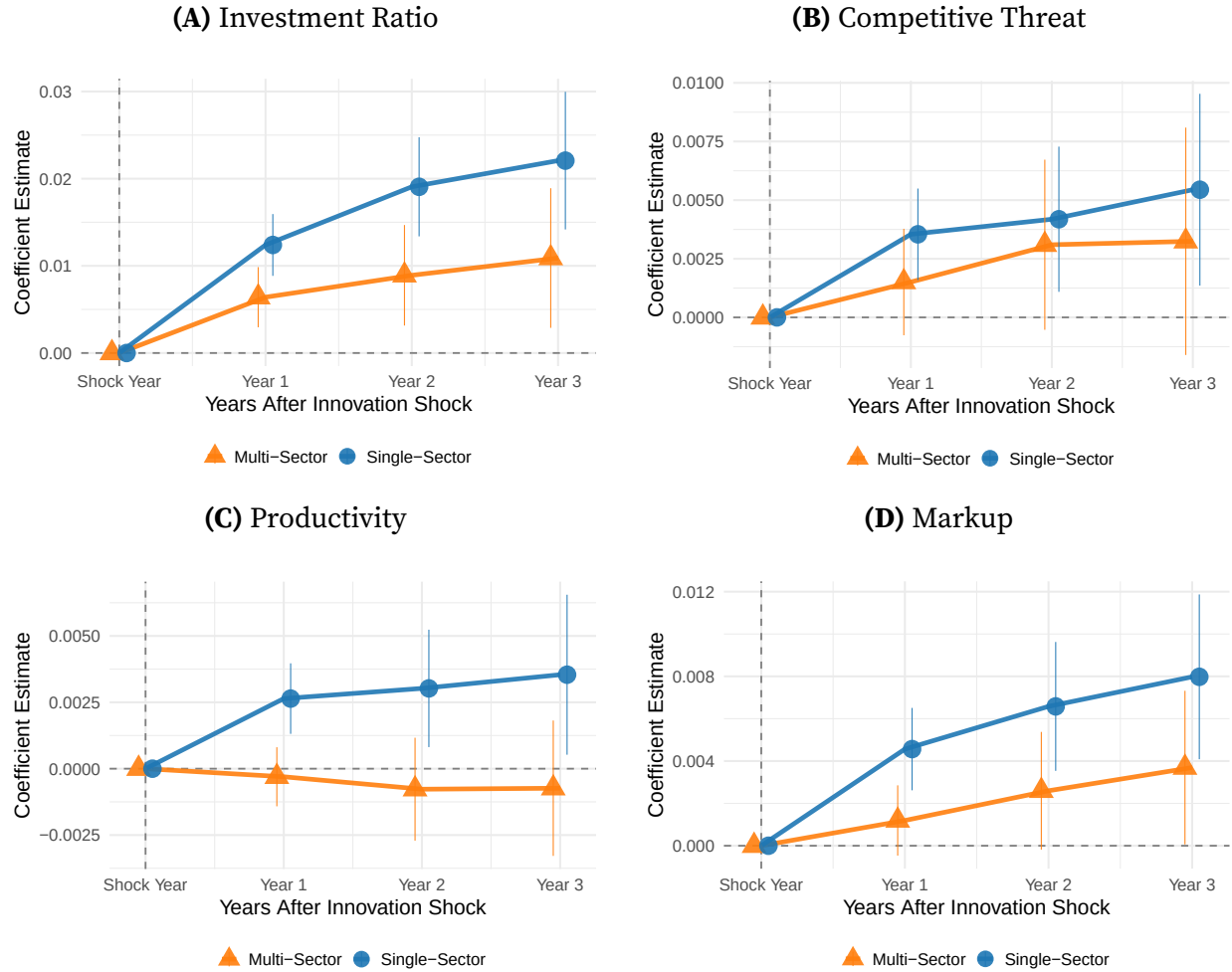


Figure 3. Dynamic Responses to Innovation Shocks by Distant Scope (Compustat Business Segments)

Note: The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (3), separately for two groups defined by distant scope. *Single-Sector* firms (low distant scope, focused) report one Compustat business segment in the year prior to the shock; *Multi-Sector* firms (high distant scope, diversified) report two or more. The shock enters as $\log(1 + V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h -year log change, $\Delta_h \log(Y)$, and all controls enter in logs; productivity and markup use the Olley-Pakes baseline. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry-year level.

Figure 3 contrasts single-segment firms (low distant scope) with multi-segment firms (high distant scope). Responses build with a lag: neither group moves in the shock year, the two groups separate from the first year onward, and the separation widens through the three-year horizon. Single-segment innovators reallocate their intangible portfolio sharply toward R&D, with an investment-ratio response roughly twice the multi-segment

response by year three (Panel A). They draw rivals into their product space, with competitive threat rising steadily and significantly, whereas the multi-segment response is positive but imprecise, with confidence bands that include zero at most horizons (Panel B). They convert the shock into real gains on both remaining margins: productivity rises significantly and persistently (Panel C), and markups climb steadily over the horizon (Panel D). The multi-segment pattern across these panels is the more revealing one. Multi-segment innovators are not inert. They shift investment toward R&D at roughly half the single-segment rate, and their markups rise by roughly half the single-segment response at year three. What they never do is become more productive: the multi-segment productivity response is flat to mildly negative at every horizon and statistically indistinguishable from zero (Panel C). The diversified firm thus converts innovation into margin without converting it into efficiency, which is the signature of the friction regime. The rents that survive fragmentation are those that brand and organisational capital can defend, not those that synergistic deployment creates.

Figure 4 reports the parallel exercise for related scope, contrasting firms above and below the sample median of *d2vscope*. Here the weak group does not merely respond less. It responds in the opposite direction. Broad-related innovators tilt their intangible investment toward R&D roughly two and a half times as strongly as narrow-related innovators by year three (Panel A), attract a significant and growing rise in competitive threat (Panel B), and secure persistent gains in productivity and markups (Panels C and D). Narrow-related innovators experience the mirror image. Rivals retreat from their product space, with the fluidity response significantly negative in the first year and recovering to zero only at the end of the horizon (Panel B). Their productivity turns negative and stays negative through year three (Panel C), consistent with these firms bearing the input costs of R&D without the deployment outlets across which its returns compound: innovation inputs rise while output gains fail to materialise beyond a single narrow market. Their markups sit flat at zero throughout (Panel D). For a firm whose related scope is too narrow to form a portfolio, an innovation of a given market value is, on the operating margins, a net burden.

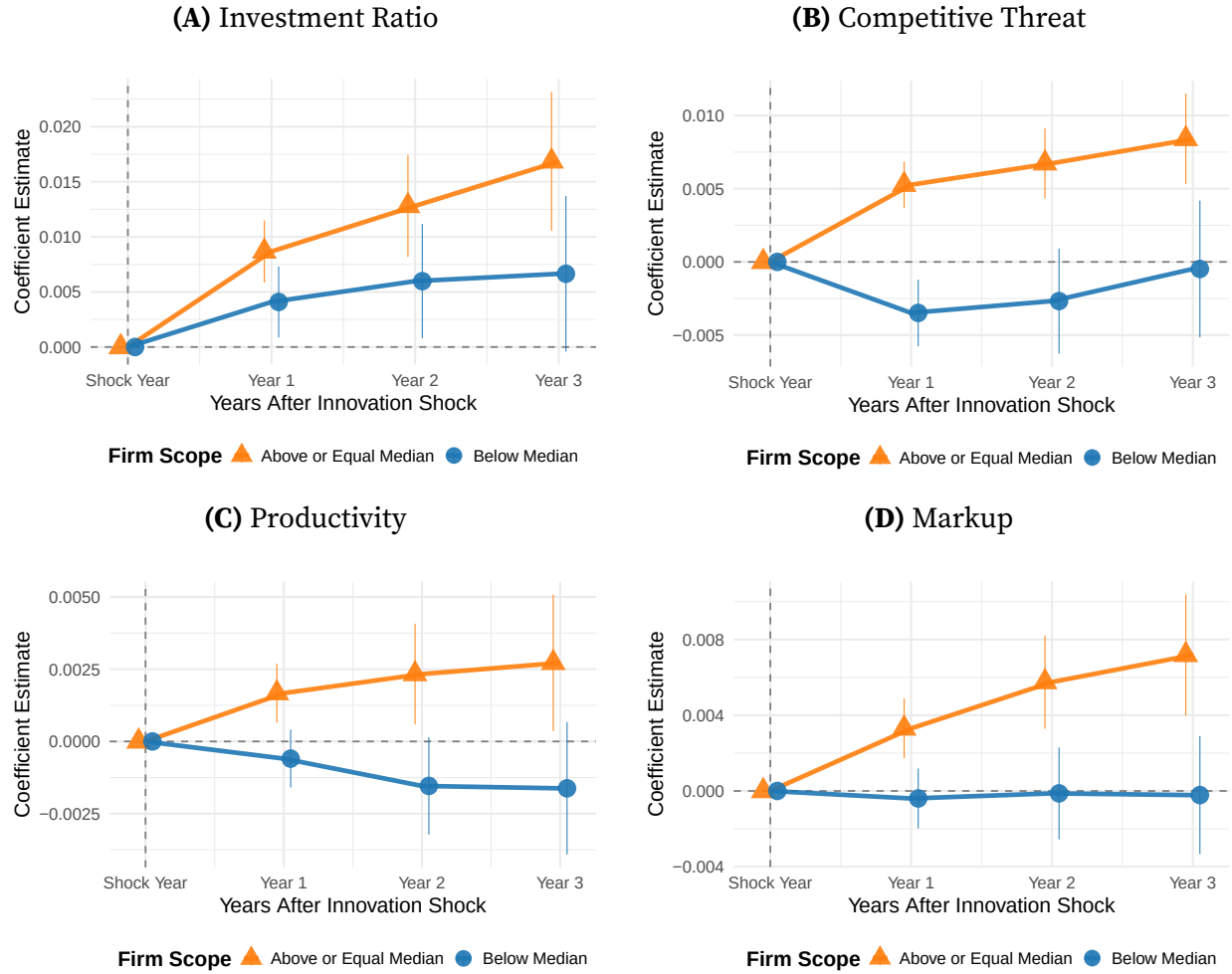


Figure 4. Dynamic Responses to Innovation Shocks by Related Scope (Hoberg–Phillips Product-Market Clusters)

Note: The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (3), separately for two groups defined by related scope. *Above or Equal Median* firms (broad related scope) are firm-years whose Hoberg–Phillips $d2vs$ scope lies at or above the sample median in the year prior to the shock; *Below Median* firms (narrow related scope) lie below the median. The shock enters as $\log(1 + V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h -year log change, $\Delta_h \log(Y)$, and all controls enter in logs; productivity and markup use the Olley–Pakes baseline. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry–year level.

Read together, the two figures deliver the paper’s central result, and the comparison between them sharpens the mechanism. Distant scope *attenuates* the conversion of innovation into rents: multi-segment innovators respond at roughly half strength on investment composition and markups, and not at all on productivity. Insufficient related scope *inverts* it: narrow-related innovators repel rivals and lose productivity outright. Productivity is

the margin on which the regimes separate most cleanly. It rises persistently wherever the firm commands synergistic outlets, through focused operations or a broad related portfolio, and it is zero or negative wherever it does not, exactly as the deployment logic predicts. The combination of rising markups with flat productivity among diversified innovators points in the same direction: appropriation through entrenchment rather than through efficiency, the hallmark of headquarters capital. The same well-identified innovation event is thus converted into contested, productivity-backed market leadership by one scope structure and dissipated or defensively absorbed by the other, and no scalar measure of scope could distinguish the two cases.

5. Conclusion

This paper has shown that the relationship between firm scope and firm performance is not a scalar fact about how many markets a firm serves but a structural fact about how its scope is composed. Related scope, the closely related sub-markets a firm serves within the boundary of a shared technological base, and distant scope, the operationally distinct domains it spans beyond that boundary, are separately measurable, and in the data they describe different margins of expansion. The typical firm is narrow across distant domains yet spreads broadly across related product markets, so collapsing the two dimensions into a single count discards precisely the variation that carries the economics.

The dynamic evidence shows that the two dimensions carry opposite economics. After a patent-value innovation shock, firms whose structure supplies synergistic outlets, single-segment firms and firms with broad related scope, tilt intangible investment toward R&D, draw rivals into their product space, and convert the shock into persistent productivity and markup gains. Firms whose structure is dominated by friction do not. Distant scope attenuates the response: multi-segment innovators reallocate investment and raise markups at roughly half strength, yet their productivity never moves, so innovation buys margin without efficiency, the signature of rents defended by brand and organisational capital rather than created by deployment. Insufficient related scope inverts the

response: narrow-related innovators repel rivals and lose productivity outright, bearing the input costs of research without the portfolio across which its returns compound.

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Appendices

A. Datasets

Table A1. Firm Scope: Compustat vs. Hoberg-Phillips

Company	Scope (Compustat)	Scope (Hoberg-Phillips)
TESLA INC	Energy Generation & Storage Automotive	Batteries
		Automotive (Brakes, Trim, Axle, Engines, Chassis)
		Automotive Safety (Airbags)
		Car Dealerships
		Energy / Cogeneration
		Utilities / Electric Power
		Smart Metering / Grid Tech
		Power Electronics / Voltage
		Solar / Renewable Energy
		Hardware & Software Solutions
		Ticketing / Scanning (Software/Systems)

B. Additional Figures and Empirical Results

Table A2. Regression Results: Compustat

Panel A: Markup and Productivity

	Pooled OLS		Two-way FE	
	(1) <i>Markup</i>	(2) <i>Productivity</i>	(3) <i>Markup</i>	(4) <i>Productivity</i>
Segment	−0.058*** (0.005)	−0.026** (0.009)	−0.070*** (0.005)	−0.005 (0.004)
Controls	Yes	Yes	Yes	Yes
Num. Obs.	33,376	33,376	33,376	33,376
Adj. R^2	0.045	0.027	0.091	0.759
RMSE	0.53	0.79	0.51	0.39
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Panel B: Investment Ratio and Competitive Threat

	Pooled OLS		Two-way FE	
	(5) <i>RD/XSGA</i>	(6) <i>Comp. Threat</i>	(7) <i>RD/XSGA</i>	(8) <i>Comp. Threat</i>
Segment	−0.117*** (0.014)	−0.461*** (0.032)	−0.115*** (0.015)	−0.313*** (0.034)
Controls	Yes	Yes	Yes	Yes
Num. Obs.	35,824	30,097	35,824	30,097
Adj. R^2	0.056	0.042	0.146	0.169
RMSE	1.29	3.04	1.22	2.83
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors clustered by *firm id* (gvkey) are in parentheses. Significance levels: $^+p < 0.1$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Controls include *sale*, *at* (assets), *xrd* (R&D), and *xsga* (SG&A). Columns 1–2 and 5–6: Pooled OLS specifications; Columns 3–4 and 7–8: Two-way fixed effects (year and 2-digit NAICS industry).

Table A3. Regression Results: HP Scope

Panel A: Markup and Productivity

	Pooled OLS		Two-way FE	
	(2) <i>Markup</i>	(3) <i>Productivity</i>	(6) <i>Markup</i>	(7) <i>Productivity</i>
HP-Scope	0.010*** (0.001)	0.023*** (0.002)	0.007*** (0.001)	0.012*** (0.001)
Controls	Yes	Yes	Yes	Yes
Num. Obs.	52,323	52,323	52,323	52,323
Adj. R^2	0.036	0.050	0.070	0.779
RMSE	0.52	0.77	0.51	0.37
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Panel B: Investment Ratio and Competitive Threat

	Pooled OLS		Two-way FE	
	(1) <i>RD/XSGA</i>	(4) <i>Comp. Threat</i>	(5) <i>RD/XSGA</i>	(8) <i>Comp. Threat</i>
HP-Scope	0.039*** (0.002)	0.201*** (0.006)	0.037*** (0.002)	0.248*** (0.006)
Controls	Yes	Yes	Yes	Yes
Num. Obs.	56,484	50,285	56,484	50,285
Adj. R^2	0.074	0.158	0.152	0.339
RMSE	1.26	2.94	1.21	2.60
FE: <i>Year</i>	No	No	Yes	Yes
FE: <i>Industry</i>	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors clustered by *firm id* (gvkey) are in parentheses. Significance levels: $^+p < 0.1$, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Controls include *sale*, *at* (assets), *xrd* (R&D), and *xsga* (SG&A). Columns (1)–(4): Pooled OLS specifications; Columns (5)–(8): Two-way fixed effects (year and 2-digit NAICS industry).

Table A4. Summary Statistics

Panel A: Compustat–Segment Merged						
Variable	N	Mean	SD	P25	Median	P75
Sale	56,734	2,075,346	11,681,302	21,007	104,878	636,445
Emp	56,734	6,064	24,512	126	510	2,800
AT	56,734	2,745,416	18,470,719	24,663	120,218	718,334
XRD	56,734	102,283	631,498	1,461	6,707	29,709
XSGA	56,734	415,856	2,024,726	10,964	39,256	166,198
Productivity	52,323	19.871	23.355	7.736	10.887	22.931
Markup	51,824	1.537	1.026	0.988	1.214	1.665
Competitive Threat	50,285	6.769	3.203	4.437	6.365	8.603

Panel B: Hoberg–Phillips Scope Merged						
Variable	N	Mean	SD	P25	Median	P75
Sale	35,964	2,171,465	10,468,548	25,104	135,367	826,899
Emp	35,964	6,440	23,946	133	571	3,212
AT	35,964	3,083,646	19,749,630	30,984	167,030	977,718
XRD	35,964	127,011	736,443	1,813	8,868	39,625
XSGA	35,964	476,449	2,256,258	13,128	50,168	208,927
Productivity	33,376	20.553	24.627	7.888	11.079	23.674
Markup	33,035	1.545	1.048	0.986	1.207	1.673
Competitive Threat	30,097	6.451	3.104	4.189	6.005	8.156

Notes: This table reports summary statistics for the two estimation samples. Panel A shows Compustat Fundamentals merged with the Compustat Segment dataset, while Panel B shows Compustat Fundamentals merged with the HP Firm Scope dataset.